

- 2 The photograph shows water lilies in a river in Thailand.



(Source: © Igor Prahin/Alamy Stock Photo)

- (a) One water lily leaf has a diameter of 305 cm.

Which of the following shows the upper surface area of this leaf?

Assume the leaf is circular.

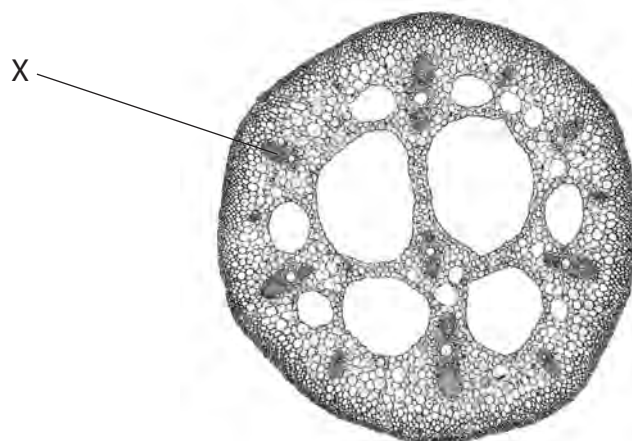
Use the formula πr^2 .

(1)

- ☐ **A** 7.3 m²
- ☐ **B** 29.22 m²
- ☐ **C** 730.6 cm²
- ☐ **D** 2922.5 cm²



(b) The photograph shows a cross-section of the stem of a water lily.



(Source: © Scenics & Science/Alamy Stock Photo)

The structure labelled X contains xylem vessels and phloem.

How many of the following statements are correct?

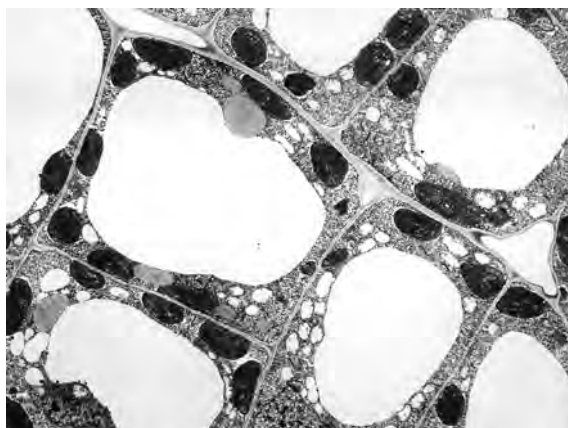
- both xylem vessels and phloem transport dissolved substances
- only phloem transport dissolved substances in one direction
- only the walls of xylem vessels contain both cellulose and lignin

(1)

- ☐ **A** none
- ☐ **B** one
- ☐ **C** two
- ☐ **D** three

- (c) The leaves of the water lilies contain several types of tissues and cells.

The photograph shows a group of leaf mesophyll cells, as seen using an electron microscope.



Magnification $\times 7\,800$

(Source: © DR. JEREMY BURGESS/SCIENCE PHOTO LIBRARY)

- (i) Label one permanent vacuole on this photograph.

(1)

- (ii) Give two functions of a vacuole.

(2)

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(Total for Question 2 = 5 marks)

